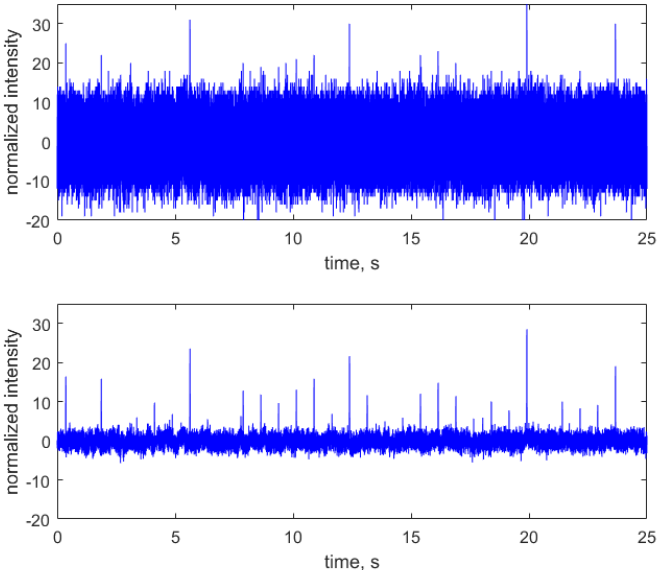
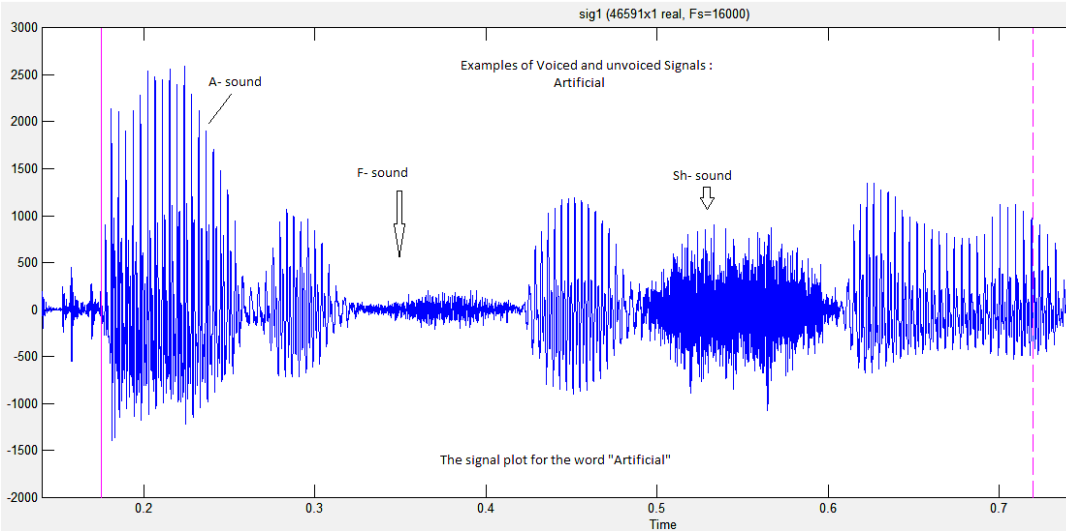
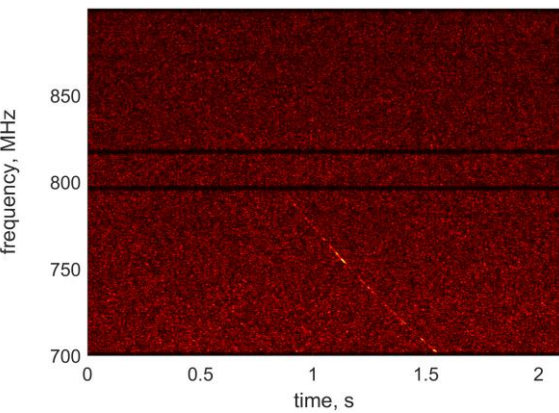
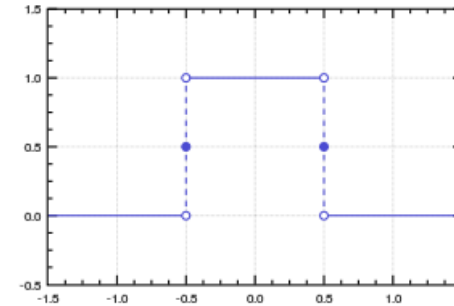




Periodic and non-periodic signals

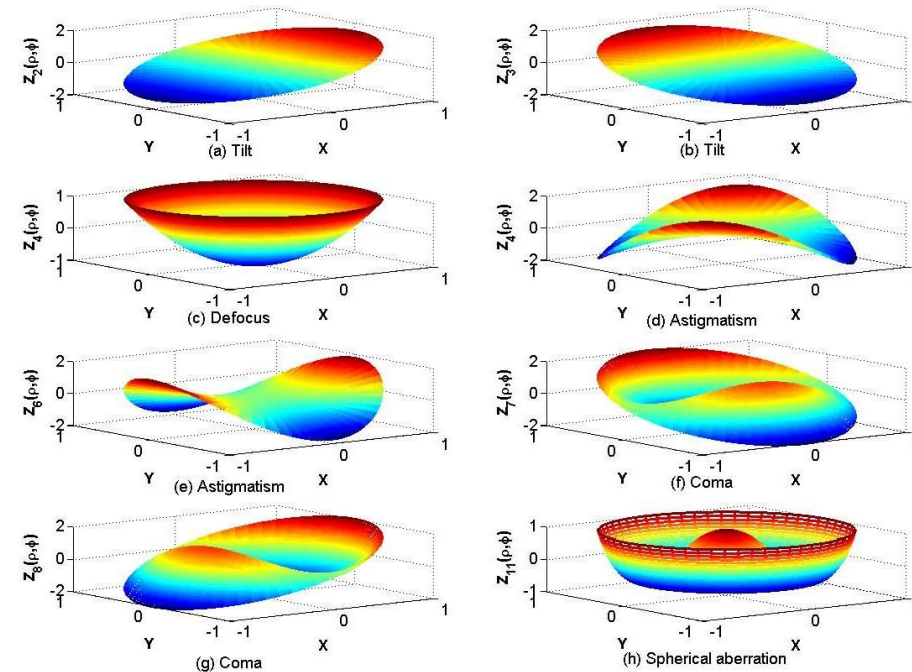
Signal decomposition

- A signal can be represented as a combination of other signals
- Example (last time)
- $\Lambda(0.5t) - \Lambda(t)$
- Depending on a type of signal, it can be decomposed in a series
- $x(t) = \sum_{k=-\infty}^{+\infty} a_k \varphi_k(t)$
- a_k are weights,
- $\varphi_k(t)$ are known functions.



Gate Functions

Zernike polynomials



Trigonometric Series

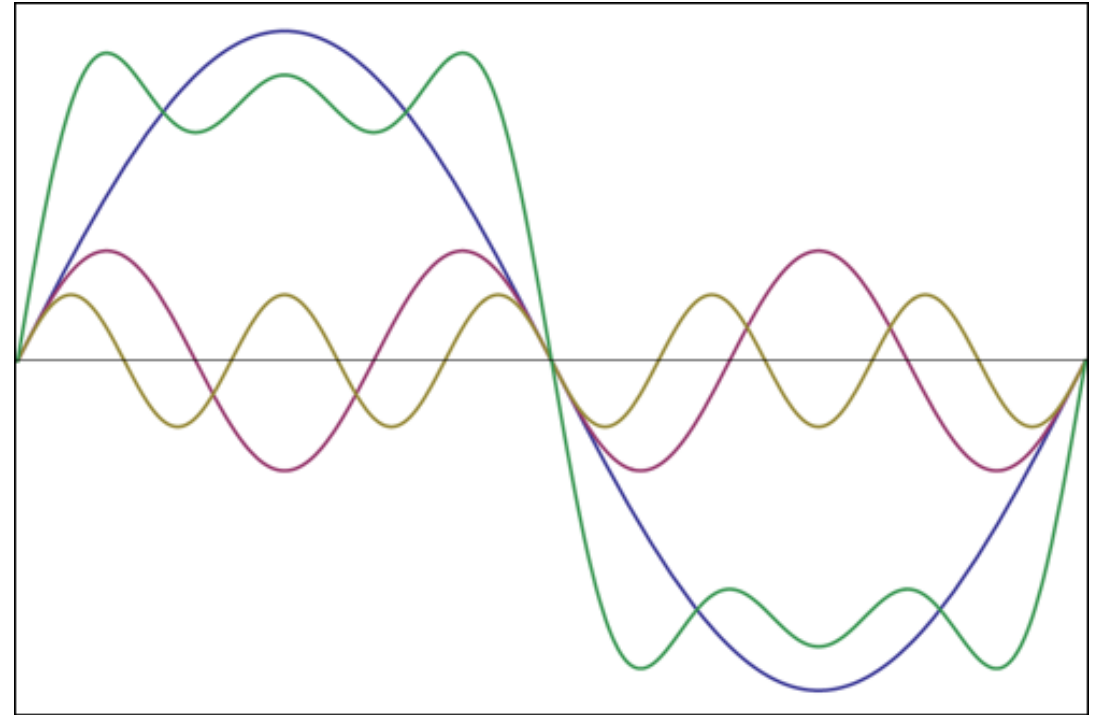
- A periodic signal can be represented as a sum of sine and cosine functions on a single period:

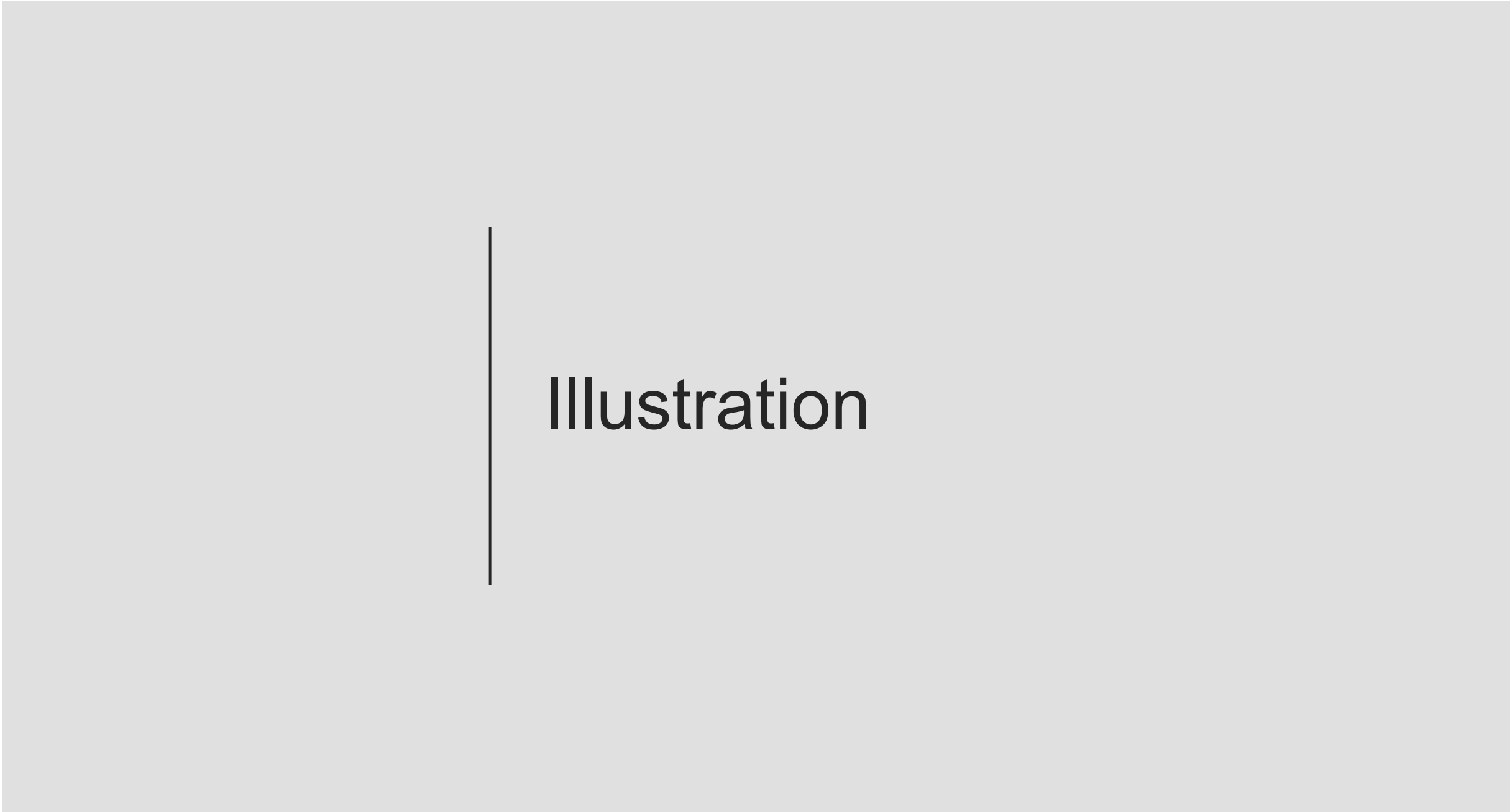
$$x(t) = b_0 + \sum_{k=1}^{+\infty} b_k \cos(\omega_0 k t) + \sum_{k=1}^{\infty} c_k \sin(\omega_0 k t)$$

$$b_k = \frac{2}{T_0} \int_{t_0}^{t_0+T_0} x(t) \cos(\omega_0 k t) dt$$

$$c_k = \frac{2}{T_0} \int_{t_0}^{t_0+T_0} x(t) \sin(\omega_0 k t) dt$$

$$b_0 = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) dt$$





Illustration

Complex Exponential Series: Background

- In place of sine and cosine functions we can use a series of complex exponentials.

- Recall Euler's equations:

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

$$e^{-j\omega t} = \cos(\omega t) - j \sin(\omega t)$$

- From these two equations:

$$\cos(\omega t) = \frac{1}{2} (e^{j\omega t} + e^{-j\omega t})$$

$$\sin(\omega t) = \frac{1}{2j} (e^{j\omega t} - e^{-j\omega t})$$

- Also,

$$\cos(\omega t) = 2\operatorname{Re}\{e^{j\omega t}\}$$

$$\sin(\omega t) = -2\operatorname{Im}\{e^{j\omega t}\}$$

- If displayed in a complex plane, $e^{j\omega t}$ is a rotating phasor

- For illustration, see

<http://spfirst.gatech.edu/chapters/02sines/demos/phasors/index.html>

Complex Exponential Series

- On a single period a real values function can be represented as a series of complex exponential functions

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{j\omega_0 kt}$$

$$a_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-j\omega_0 kt} dt$$

- a_k tell us how much the k-th discrete frequency contributes to signal's representation
- a_k , b_k , and c_k are related

$$a_0 = b_0$$

$$b_k = 2\text{Re}\{a_k\}$$

$$c_k = -2\text{Im}\{a_k\}$$
- Fast Fourier Transform (FFT) is based on complex exponentials

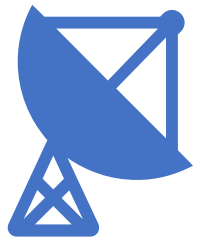


Illustration: Radio Astronomy

